

JARVIS v0.1 — Personal Research Scout

FL-06 · Design Your Personal Agent · Build Phase

Design principle: JARVIS is the long-term personal-agent concept; this assignment deliberately builds one narrow capability rather than attempting a general-purpose assistant.

1. Job to be done

JARVIS turns a research question into a concise, source-grounded research brief. It determines what evidence is needed, searches approved personal/project material first when relevant, uses external web sources when required, compares evidence, separates fact from inference, and reports uncertainty.

Out of scope: email management, calendar actions, task automation, coding assistance, voice control, smart-home control, autonomous purchasing, and general life management.

2. User and usage

Primary user: Me — the owner of JARVIS and its project/learning materials.

Frequency: As needed during project work and learning; target several research tasks per week.

Typical input: A question, decision, topic, or claim that needs evidence.

Output: A short research brief with answer, key findings, evidence/source list, uncertainty, and next step.

3. Tools, data, and access plan

Tool / data	Purpose	Access plan	Boundary
Personal/project files	Ground research in the user's own notes, project docs, and assignment material.	Local project folder / approved document retrieval available to the scripted agent.	Use when the question refers to personal material; do not invent missing content.
Web search	Find current or external evidence when personal files are insufficient.	Search API/tool available to the scripted agent; record source title/URL/date where available.	Do not claim a search occurred if the tool failed or was not used.
LLM / agent runtime	Interpret the request, choose tool calls, synthesize evidence, and format the brief.	Scripted agent using an accessible model/API or local model available in the build environment.	Model knowledge is not treated as verified research evidence.
Optional file output	Save the final brief for reuse.	Write only to a designated JARVIS output folder.	No deletion/overwrite of unrelated files without explicit confirmation.

4. Draft agent instructions

- Act as JARVIS, a personal research scout. Your job is to produce reliable, concise research briefs — not to act as a general personal assistant.
- First determine whether the request is clear enough to research. If a missing detail could materially change the result, ask one focused clarification before searching.
- When the request concerns the user's own files or project decisions, search those approved sources first. Treat retrieved material as the source of truth for claims about the user's work.
- Use external web research when the question requires current, external, or corroborating information. Prefer authoritative and primary sources where practical.
- Track provenance: distinguish user-provided evidence, external evidence, model inference, and unresolved uncertainty.
- Never fabricate sources, quotations, search results, or file contents. If evidence is insufficient or conflicting, say so explicitly.
- Return a compact brief: Question → Short answer → Key findings → Evidence/sources → Uncertainty or conflicts → Recommended next step.
- Do not take irreversible external actions. If a future capability would send, delete, publish, purchase, or materially modify something, require explicit user confirmation first.

5. Pre-build evaluation cases

# / scenario	Expected behavior	Pass condition	Failure signal
1. Straight research question	Researches and produces a concise brief with evidence.	Answer is supported and sources are clearly identified.	Unsupported factual claims or missing provenance.

# / scenario	Expected behavior	Pass condition	Failure signal
2. Multi-source synthesis	Combines multiple relevant sources without collapsing them into one unsupported claim.	Sources agree or differences are explained.	Cherry-picking or invented synthesis.
3. Conflicting sources	Flags the conflict and explains why the evidence differs.	Conflict is visible; JARVIS does not pretend certainty.	Picks a winner silently.
4. Insufficient evidence	States that evidence is insufficient and asks for a narrower question or more material.	No hallucinated answer/citation.	Fills gaps with confident speculation.
5. Ambiguous request	Asks one targeted clarification when ambiguity materially changes the research.	Clarification precedes unnecessary tool use.	Guesses the user's intent.
6. Personal-file vs web conflict	Prioritizes personal material for claims about the user's project, while noting external disagreement.	Both provenance and conflict are explicit.	Rewrites the user's project history from web assumptions.

6. Risks and guardrails

Hallucinated evidence: Must never fabricate sources, citations, quotes, or retrieved file content.

False certainty: Must label uncertainty and distinguish fact from inference.

Wrong source priority: For personal/project questions, use approved user material first and preserve its provenance.

Ambiguous intent: Ask for clarification when ambiguity could change the result materially.

Irreversible actions: No sending, publishing, deleting, purchasing, or consequential modification without explicit confirmation.

Privacy: Use only approved user-provided files; do not expose private material in outputs intended for external sharing.

7. Platform choice

Chosen: scripted agent on the scripting path. The prototype can implement a small, explicit tool loop without committing the project to a broad automation platform. The design requires only document retrieval, web search, model reasoning, and optional file output. A scripted prototype also makes the six evals reproducible.

Alternative: a Claude Project with connectors and skills would be faster for document-grounded prototyping, but introduces a platform/access dependency and provides less direct control over the evaluation harness. The scripted path is preferable if the required model/API or local model is already available; otherwise switch rather than spending the assignment building infrastructure.

8. Build boundary and success criterion

Target build time: approximately 8–10 hours. Version 0.1 is successful if it reliably completes the six evaluation cases, produces source-grounded briefs, and refuses or requests confirmation for the defined risky situations. Anything outside the research-scout workflow is deferred to future JARVIS versions.

The portfolio identity is deliberately not part of the agent's functional scope. JARVIS may later share the same visual system, but FL-06 is evaluated on agent design, tools, evals, and guardrails.